# EXPERIMENT - 4

## Aim:-

Using the Employee schema, demonstrate:

- `INNER JOIN`
- `LEFT JOIN`
- Self Join
- 3-way Join
- Correlated Subquery
- `EXISTS`
- Simulated `INTERSECT`
- Simulated `EXCEPT`
- `EXPLAIN` execution plans

## Source Code:-

**INNER JOIN**

```
SELECT
   e.emp_name,
   d.dept_name
FROM Employee e
INNER JOIN Department d
   ON e.dept_id = d.dept_id;
```

**LEFT JOIN**

```
SELECT
   d.dept_name,
   e.emp_name
FROM Department d
LEFT JOIN Employee e
   ON d.dept_id = e.dept_id;
```

**SELF JOIN**

```
SELECT
   e1.emp_name AS employee_1,
   e2.emp_name AS employee_2,
   e1.dept_id
FROM Employee e1
JOIN Employee e2
   ON e1.dept_id = e2.dept_id
   AND e1.emp_id < e2.emp_id;
```

**3-WAY JOIN**

```
SELECT
   e.emp_name,
   d.dept_name,
   p.project_name,
   e.salary
FROM Employee e
JOIN Department d
   ON e.dept_id = d.dept_id
JOIN Project p
   ON e.project_id = p.project_id;
```

**CORRELATED SUBQUERY**

```
SELECT
   e.emp_name,
   e.salary,
   e.dept_id
FROM Employee e
WHERE e.salary >
(
   SELECT AVG(e2.salary)
   FROM Employee e2
   WHERE e2.dept_id = e.dept_id
);
```

**EXISTS**

```
SELECT
   d.dept_id,
   d.dept_name
FROM Department d
WHERE EXISTS
(
   SELECT 1
   FROM Employee e
   WHERE e.dept_id = d.dept_id
     AND e.salary > 90000
);
```

**SIMULATED INTERSECT**

```
SELECT e.emp_id, e.emp_name
FROM Employee e
WHERE e.dept_id = 2

AND e.emp_id IN
(
   SELECT emp_id
   FROM Employee
   WHERE salary > 70000
);
```

**SIMULATED EXCEPT**

```
SELECT emp_id, emp_name
FROM Employee
WHERE dept_id = 2

AND emp_id NOT IN
(
   SELECT emp_id
   FROM Employee
   WHERE salary > 70000
);
```

**EXPLAIN - INNER JOIN**

```
EXPLAIN
SELECT
   e.emp_name,
   d.dept_name
FROM Employee e
INNER JOIN Department d
   ON e.dept_id = d.dept_id;
```

**EXPLAIN - LEFT JOIN**

```
EXPLAIN
SELECT
   d.dept_name,
   e.emp_name
FROM Department d
LEFT JOIN Employee e
   ON d.dept_id = e.dept_id;
```

# Output:-

## INNER JOIN

| emp_name | dept_name |
|---|---|
| Aarav Sharma | Computer Science Engineering |
| Rohan Verma | Computer Science Engineering |
| Isha Patel | Computer Science Engineering |
| Arjun Das | Computer Science Engineering |
| Nikhil Jain | Computer Science Engineering |
| Tanya Roy | Computer Science Engineering |
| Dev Thakur | Computer Science Engineering |
| Ananya Singh | Sciences |
| Rahul Joshi | Sciences |
| Vikram Rao | Sciences |
| Simran Kaur | Sciences |
| Yash Agarwal | Sciences |
| Nandini Iyer | Sciences |
| Priya Mehta | Law |
| Neha Kapoor | Law |

## LEFT JOIN

| dept_name | emp_name |
|---|---|
| Computer Science Engineering | Aarav Sharma |
| Computer Science Engineering | Rohan Verma |
| Computer Science Engineering | Isha Patel |
| Computer Science Engineering | Arjun Das |
| Computer Science Engineering | Nikhil Jain |
| Computer Science Engineering | Tanya Roy |
| Computer Science Engineering | Dev Thakur |
| Sciences | Ananya Singh |
| Sciences | Rahul Joshi |
| Sciences | Vikram Rao |
| Sciences | Simran Kaur |
| Sciences | Yash Agarwal |
| Sciences | Nandini Iyer |
| Law | Priya Mehta |
| Law | Neha Kapoor |

# SELF JOIN

| employee_1 | employee_2 | dept_id |
|---|---|---|
| Aarav Sharma | Rohan Verma | 1 |
| Aarav Sharma | Isha Patel | 1 |
| Aarav Sharma | Arjun Das | 1 |
| Aarav Sharma | Nikhil Jain | 1 |
| Aarav Sharma | Tanya Roy | 1 |
| Aarav Sharma | Dev Thakur | 1 |
| Ananya Singh | Rahul Joshi | 2 |
| Ananya Singh | Vikram Rao | 2 |
| Ananya Singh | Simran Kaur | 2 |
| Ananya Singh | Yash Agarwal | 2 |
| Ananya Singh | Nandini Iyer | 2 |
| Rohan Verma | Isha Patel | 1 |
| Rohan Verma | Arjun Das | 1 |
| Rohan Verma | Nikhil Jain | 1 |
| Rohan Verma | Tanya Roy | 1 |

# 3-WAY JOIN

| emp_name | dept_name | project_name | salary |
|---|---|---|---|
| Aarav Sharma | Computer Science Engineering | Campus Management System | 65000.00 |
| Rohan Verma | Computer Science Engineering | Cyber Security Platform | 58000.00 |
| Isha Patel | Computer Science Engineering | Campus Management System | 61000.00 |
| Arjun Das | Computer Science Engineering | Cyber Security Platform | 63000.00 |
| Nikhil Jain | Computer Science Engineering | Campus Management System | 66000.00 |
| Tanya Roy | Computer Science Engineering | Cyber Security Platform | 62000.00 |
| Dev Thakur | Computer Science Engineering | Campus Management System | 64000.00 |
| Ananya Singh | Sciences | AI Chatbot | 72000.00 |
| Rahul Joshi | Sciences | Computer Vision System | 92000.00 |
| Vikram Rao | Sciences | AI Chatbot | 88000.00 |
| Simran Kaur | Sciences | Computer Vision System | 73000.00 |
| Yash Agarwal | Sciences | Computer Vision System | 95000.00 |
| Nandini Iyer | Sciences | AI Chatbot | 83000.00 |
| Priya Mehta | Law | Employee Analytics | 81000.00 |
| Neha Kapoor | Law | Payroll Automation | 69000.00 |

# CORRELATED SUBQUERY

| emp_name | salary | dept_id |
|---|---|---|
| Aarav Sharma | 65000.00 | 1 |
| Priya Mehta | 81000.00 | 3 |
| Sneha Gupta | 68000.00 | 5 |
| Rahul Joshi | 92000.00 | 2 |
| Karan Malhotra | 75000.00 | 4 |
| Vikram Rao | 88000.00 | 2 |
| Arjun Das | 63000.00 | 1 |
| Kavya Shah | 79000.00 | 3 |
| Nikhil Jain | 66000.00 | 1 |
| Varun Sethi | 85000.00 | 4 |
| Yash Agarwal | 95000.00 | 2 |
| Mohit Bansal | 77000.00 | 4 |
| Aditi Chawla | 68000.00 | 5 |
| Dev Thakur | 64000.00 | 1 |
| Shreya Bose | 74000.00 | 3 |

# EXISTS

| dept_id | dept_name |
|---|---|
| 2 | Sciences |

# SIMULATED INTERSECT

| emp_id | emp_name |
|---|---|
| 2 | Ananya Singh |
| 7 | Rahul Joshi |
| 11 | Vikram Rao |
| 16 | Simran Kaur |
| 21 | Yash Agarwal |
| 26 | Nandini Iyer |

## SIMULATED EXCEPT

No data - please run the query to get the data

## EXPLAIN - INNER JOIN

| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SIMPLE | d | null | ALL | PRIMARY | null | null | null | 5 | 100 | null |
| 1 | SIMPLE | e | null | ref | dept_id | dept_id | 5 | db_453k9p6f4.d.dept_id | 6 | 100 | null |

## EXPLAIN - LEFT JOIN

| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SIMPLE | d | null | ALL | null | null | null | null | 5 | 100 | null |
| 1 | SIMPLE | e | null | ref | dept_id | dept_id | 5 | db_453k9p6f4.d.dept_id | 6 | 100 | null |